P8X Backplane Bus -- Quick Reference (rev C)

DIN 41612 96-pin pin→signal map (component-side view of card connector)

Pin	Row A	Row B	Row C
1	+5V	+5V	+5V
2	+5V	+5V	+5V
3	D0	GND	A0
4	D1	SPARE12	A1
5	D2	GND	A2
6	D3	SPARE13	A3
7	D4	GND	A4
8	D5	SPARE14	A5
9	D6	GND	A6
10	D7	SPARE15	A7
11	-RES	GND	A8
12	DOE0	SPARE16	A9
13	DOE1	GND	A10
14	DOE2	SPARE17	A11
15	DOE3	GND	A12
16	DLD0	SPARE18	A13
17	DLD1	GND	A14
18	DLD2	SPARE19	A15
19	DLD3	GND	ALUS0
20	PSEL0	SPARE20	ALUS1
21	PSEL1	GND	ALUS2
22	PINC	SPARE21	ALUS3
23	PDEC	GND	ALUM
24	CLK	SPARE22	CIN
25	CLKB	GND	SH0
26	LDF	SPARE23	SH1
27	FC	CLRC	PSEL2
28	FZ	BSEL	LDZN
29	FN	IRQ	SHCIN
30	FV	SPARE11	SETC
31	GND	GND	GND
32	GND	GND	GND

Signal Groups

Group	Signals	Meaning
Data bus	D0-D7 (A3-A10)	8-bit bidirectional data bus; one driver/microcycle (one-hot DOE).
Address bus	A0-A15 (C3-C18)	16-bit address; driven solely by the register-bank card (selected pointer).
DOE field	DOE0-3 (A12-A15)	Data Output Enable: selects which card drives the data bus.
DLD field	DLD0-3 (A16-A19)	Data LoaD: selects which register latches the data bus.
Pointer sel	PSEL0-1 (A20-A21), PSEL2 (C27)	Selects active pointer P0-P3 (+PT scratch=4); PINC/PDEC inc/dec it.
Ptr inc/dec	PINC (A22), PDEC (A23)	Increment / decrement the selected pointer.
ALU select	ALUS0-3 (C19-C22), ALUM (C23), CIN (C24)	74181 function select, mode (logic/arith), carry-in.
Shifter	SH0 (C25), SH1 (C26), SHCIN (C29)	Shifter control; SHCIN = shift-in is C flag (rotate-through-carry).
Flags out	FC FZ FN FV (A27-A30)	Carry/Zero/Negative/oVerflow; ALU card -> control card cond mux.
Flag latch	LDF (A26), LDZN (C28)	LDF latches C/Z/N/V; LDZN latches only Z,N on loads.
C control	SETC (C30), CLRC (B27)	Force C=1 (SEC) / C=0 (CLC), leaving Z/N/V untouched.
Clock/reset	CLK (A24), CLKB (A25), -RES (A11)	System clock + complement; reset active-low. (Control card -> all.)
Rev-C lines	BSEL (B28), IRQ (B29)	BSEL: ALU B-mux 0=B reg,1=T reg. IRQ: maskable int request (reserved).
Power/GND	+5V (1,2 all rows), GND (31,32 + row-B guard)	+5V/GND planes; row B 3-26 is a grounded guard between signal rows.
Spare	SPARE4-11 (rev-C reallocations noted)	Reserved, bused to all 10 slots. SPARE0-3 became FC/FZ/FN/FV.

Connector: DIN 41612, 3 rows (A/B/C) x 32 pins, 10 slots @ 25.4 mm. Shading: +5V GND spare. Generated from the same busnet() pin-map as the CAD files -- cannot drift from the boards.